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BELGAUM, KARNATAKA

Project Report on

REAL TIME INCUBATOR MONITORING SYSTEM USING WIRELESS NETWORKS

Submitted in partial fulfillment of the requirements for the award of degree

BACHELOR OF ENGINEERING
In
ELECTRONICS AND COMMUNICATION ENGINEERING

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B.M.S. INSTITUTE OF TECHNOLOGY AND MANAGEMENT

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CERTIFICATE

Certified that the project work entitled **"REAL TIME INCUBATOR MONITORING SYSTEM USING WIRELESS NETWORKS"** carried out by **Mr. Mohammed Saqib (1BY20EC101), Mr. Pravin M Malashetti (1BY20EC125), Mr. Rishabh Jaiswal (1BY20EC136),** and **Ms. Swathika M (1BY20EC194)** are bonafide students of **BMS Institute of Technology and Management** in partial fulfilment for the award of **Bachelor of Engineering** in **Electronics and Communication Engineering** prescribed by **Visvesvaraya Technological University, Belagavi** during the academic year **2022-2023.** It is suggested that all corrections/suggestions indicated for internal assessment have been indicated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Signature of Guide **Dr. Rashmi N** Dept. of ECE	Signature of HOD **Dr. Jayadeva G S** Dept. of ECE	Signature of Principal **Dr. Mohan Babu G.N.** BMSIT&M

External Viva

Name of the Examiners	Signature with Date
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ABSTRACT

The care of premature infants and newborns require intensive medical attention and it is a critical aspect of neonatal health care. Traditional incubators are not advanced in technology and are not enough to make sure that the premature infants are taken care properly for their healthy growth and development and this sometimes leads to mortality of the vulnerable premature infants. The smart baby incubator here is used to prevent those problems, designed with cutting-edge technology these incubators as proposed in this paper provide best environment for the pre mature babies. These incubators use ARDUINO UNO integrated with sensors such as DHT11 for controlling temperature and humidity, camera for observing, high precision heart pulse rate detectors, maintaining optimum oxygen and many other such features. It is connected to the internet via custom made website where the parents, top doctors can monitor and control the parameters for the baby. Overall, this provides safe environment for the baby.

ACKNOWLEDGEMENT

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CHAPTER -1

INTRODUCTION

1.1 Problem Analysis

The smart baby incubator project aims to improve infant health outcomes by providing a safe and secure environment for premature babies in hospitals. Factors to consider in problem analysis include precise temperature, humidity, and oxygen control, designing a safe and sterile environment, accessibility for healthcare providers, monitoring vital signs, and integration with existing hospital infrastructure. The project team can develop a comprehensive solution that meets these needs to improve the care and outcomes for premature babies.

1.2 Objective

- Develop an easy-to-use system for NICUs.
- Cost-effective: Utilizes Arduino microcontroller and sensors.
- User-friendly: Provides remote access to live data through website.
- Optimal environment: Regulates environmental conditions for baby's growth and development.
- Real-time monitoring: Allows doctors and parents to stay informed and make timely decisions.
- Improves quality of care: Enhances effectiveness of NICU care and reduces risk of complications for premature babies.

1.3 Motivation

- The motivation for the Smart Baby Incubator Project comes from the need to address the high rates of premature births and the associated risks and complications that premature infants face.
- Premature babies require specialized care and monitoring, including precise control of temperature, humidity, and oxygen levels, to promote healthy growth and development.
- The Smart Baby Incubator project aims to provide safe, sterile, and nurturing environment that promotes the health and well-being of premature infants.
- It is also important to provide parents with real-time monitoring and remote access to the incubator system to reduce stress and anxiety while allowing them to stay connected to their baby’s care

CHAPTER -2

LITERATURE SURVEY

Literature review based on research papers

The literature survey of the Smart Baby Incubator project highlights the use of various control systems and technologies to regulate environmental conditions in neonatal incubators. Some of the relevant studies and articles include:

- "A Survey on Neonatal Incubator Monitoring System" by Sunitha K A, and Revathi Venkataraman, SRM University-AP. This study proposed the designing of an incubator with minimum emf exposure. The system is designed in cost-effective manner with the ability to measure maximum parameters and efficient technology. Non-Contact sensing of the parameters are encouraged. Incubator is designed in portable manner by considering energy efficiency and useful technologies.

- "Development of Wireless Monitoring System for Neonatal Intensive Care Unit (NCIU)" by N.S. Joshi, R.K. Kamat, P.K. Gaikwad, International Journal of Advanced Computer Research Volume-3 Number-3 Issue-11 September-2013. Overall, this paper focuses on the development and evaluation of a wireless monitoring system specifically designed for neonatal intensive care units. It presents the system architecture, wireless communication protocols, sensor nodes, central monitoring unit, and user interface. The research findings highlight the potential benefits of such a system in enhancing healthcare monitoring in NICUs.

- "Real Time Incubator Monitoring System Using Wireless Sensor Network" by Karan Kolla, Rakesha R, Tejus S, Narendra Kumar G and Alice Abraham, Dept. of Electronics & Communication, UVCE, Bangalore University, INDIA. Overall, this paper focuses on the development and evaluation of a real-time incubator monitoring system using a wireless sensor network. It presents the system architecture, wireless communication protocols, sensor nodes, central monitoring unit, and data visualization techniques. The research findings emphasize the potential advantages of such a system in enhancing neonatal care and monitoring in incubators.

- "A design of power supply for neonatal monitoring with wearable sensors", by Wei Chen, Christoph Sonntag, Freek Boesten, Sidarto Bambang Oetomo, and Loe Feijs, Department of Industrial Design, Eindhoven University of Technology, Den Dolech 2, 5612 AZ, Eindhoven, The Netherlands. Overall, this paper focuses on the design considerations, challenges, and solutions related to the power supply for neonatal monitoring with wearable sensors. It addresses aspects such as power generation, energy harvesting, storage, regulation, and safety.

CHAPTER -3

METHODOLOGY

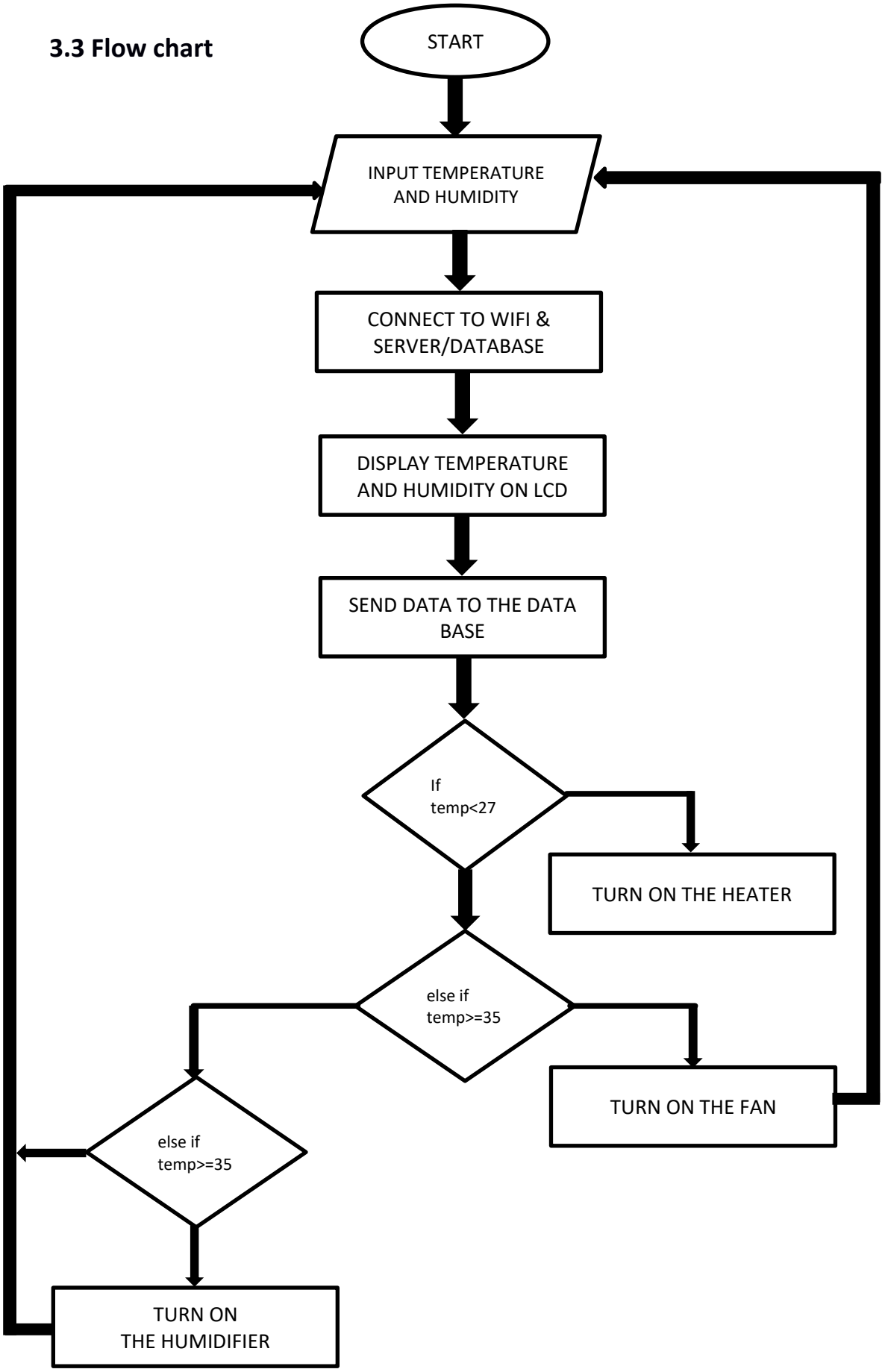
3.1 Scope of present work

The methodology of the Smart Baby Incubator project involves the development of a system that can monitor and regulate environmental conditions in a neonatal incubator. The project utilizes an Arduino-based system, which includes a microcontroller and various sensors to monitor temperature, humidity, and oxygen levels inside the incubator.

3.2 Proposed Model

- System design: Designing the system architecture, including the selection of sensors and development board.
- Sensor integration: Sensors are integrated into the system to monitor environmental conditions
- System testing: The system is tested to ensure that it can accurately monitor and regulate the environmental conditions inside the incubator.
- User interface development: A user-friendly interface is developed to display the data collected by the sensors and allow users to remotely access the system through a website.
- Integration with existing infrastructure: The system is designed to integrate with existing hospital infrastructure and protocols, including electronic health record systems.
- Live data monitoring: The system provides real-time data monitoring on the website so that doctors and parents can stay informed about the baby's condition.

3.3 Flow chart



CHAPTER -4

REAL TIME INCUBATOR MONITORING SYSTEM USING WIRELESS NETWORKS

4.1 Introduction to the project

- The healthcare industry has witnessed remarkable advancements over the years, revolutionizing the way we provide care and improve patient outcomes. One area that has seen significant innovation is neonatal care, especially in the development of baby incubators. Traditionally, incubators have played a critical role in providing a controlled environment for premature infants, ensuring their optimal growth and development. However, with the rapid advancement of technology, there is a growing need to integrate IoT (Internet of Things) capabilities into these life-saving devices.

- By incorporating IoT technology, we can enhance the functionality of incubators, enabling real-time monitoring, customized environment control, and seamless integration with electronic health records (EHR). These advancements have the potential to improve the overall quality of care provided to premature infants, while also streamlining healthcare processes for medical professionals.

- We will explore the various components and features of a smart baby incubator, including the integration of sensors to monitor vital parameters such as temperature, humidity, and oxygen levels. We will delve into the benefits of real-time monitoring, automated alerts and notifications, and the ability to customize the incubator environment based on individual baby's needs.

- Integration of incubator data with electronic health records (EHR), leading to improved accuracy, accessibility, and decision-making in neonatal care. We will also examine the safety and security considerations that come with incorporating IoT technology into these critical healthcare devices.

- By exploring case studies and success stories where smart baby incubators using IoT have been implemented, showcasing the positive impact on neonatal care and patient outcomes. We will also address the challenges and considerations associated with the adoption of these technologies, such as cost implications, privacy, and data protection.

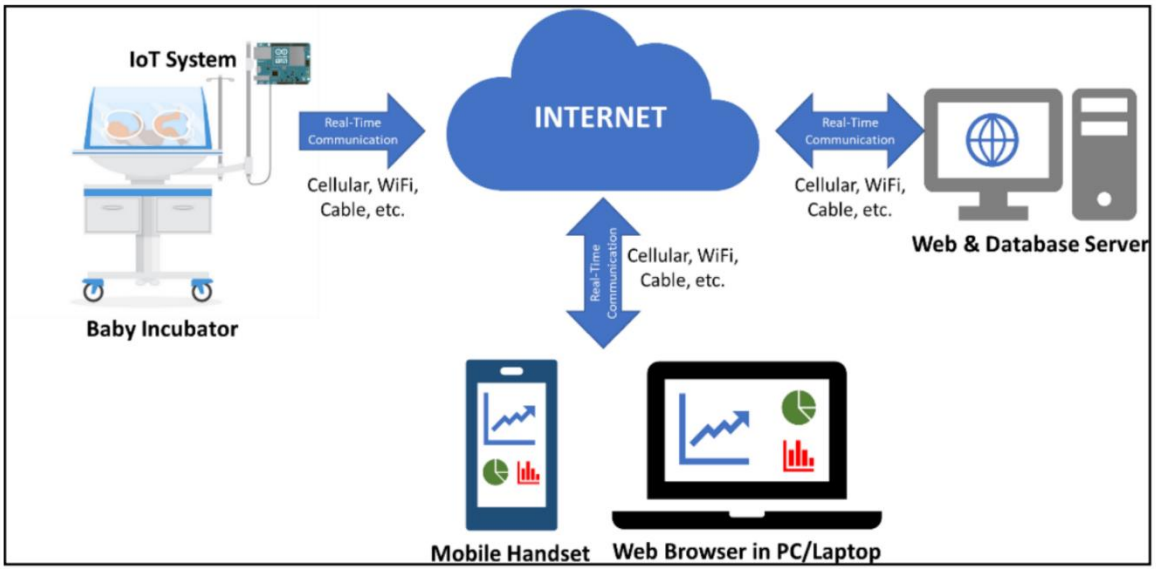


Fig 1. Project Visualization

4.2 Working

Sensors and Data Acquisition:

The smart baby incubator utilizes various sensors, such as temperature, humidity, and oxygen level sensors, to continuously monitor the vital parameters of the infant. These sensors are strategically placed within the incubator and collect real-time data on the infant's condition. For example, the temperature sensor measures the ambient temperature, ensuring it remains within the optimal range for the baby's comfort and well-being. The humidity sensor tracks the moisture levels, preventing excessive dryness or humidity. The oxygen level sensor monitors the amount of oxygen available to the baby, ensuring it remains at an appropriate level.

IoT Connectivity:

The smart baby incubator is equipped with IoT connectivity options such as Wi-Fi, Bluetooth, or cellular connectivity. This allows the incubator to establish a connection with the internet and communicate with other devices or systems. The connectivity enables seamless data transmission, remote access, and control of the incubator's settings.

Data Transmission:

The collected data from the sensors is securely transmitted over the internet to a cloud-based platform or a local server. This ensures that the data is accessible from anywhere and can be processed in real-time. Secure transmission protocols, such as encryption techniques, are employed to safeguard the privacy and integrity of the data during transmission.

Data Processing and Analysis:

The transmitted data is received and processed by the cloud-based platform or the local server. Advanced algorithms and analytics are applied to interpret the data and derive meaningful insights. The system analyzes the data to identify patterns, trends, and potential health risks. For instance, it can detect sudden changes in temperature or oxygen levels that may indicate a problem and generate alerts accordingly.

Real-time Monitoring and Alerts:

Based on the processed data, the smart baby incubator system provides real-time monitoring of the infant's vital signs. Healthcare professionals can access this information through a web or mobile application. The system continuously updates the data visualization in a user-friendly interface, allowing healthcare providers to track the infant's condition and make informed decisions promptly. If any abnormality or critical condition is detected, the system generates automated alerts or notifications. These alerts can be sent to healthcare professionals' devices, ensuring immediate attention and intervention.

Customized Environment Control:

One of the significant advantages of a smart baby incubator using IoT is the ability to remotely control the incubator's environmental factors. Healthcare professionals can access the incubator's settings through the web or mobile application. They can adjust parameters such as temperature, humidity, and oxygen levels according to the specific needs of the infant. For example, if the infant requires a higher temperature or a more humid environment, these adjustments can be made remotely, ensuring personalized care for each baby.

Integration with Electronic Health Records (EHR):

The smart baby incubator system seamlessly integrates with electronic health records (EHR) systems. The collected data from the incubator, including vital signs, environmental conditions, and alarms, can be automatically transferred to the patient's health records. This integration enhances accuracy, efficiency, and continuity of care. Healthcare professionals can access the complete history of the infant's incubation period, facilitating comprehensive assessment and informed decision-making.

Safety and Security Measures:

Given the sensitivity of the data collected by the smart baby incubator, robust safety and security measures are implemented. Encryption techniques are utilized to secure the data during transmission and storage. Access control mechanisms, such as authentication and authorization protocols, are implemented to restrict unauthorized access to the system. These measures ensure the confidentiality, integrity, and privacy of the infant's data, adhering to the strict healthcare data protection regulations.

4.3 Real-time tracking

Continuous Data Collection:

The smart baby incubator is equipped with sensors that collect real-time data on various vital signs of the infant. These may include parameters such as temperature, humidity, oxygen levels, heart rate, and respiratory rate. The sensors continuously monitor these parameters and provide a constant stream of data.

IoT Connectivity:

The incubator is connected to the internet via IoT technologies, allowing the collected data to be transmitted in real-time to a central monitoring system. This connectivity ensures seamless and instant transfer of the data, making it accessible for healthcare professionals.

Web or Mobile Application:

A web or mobile application is designed to receive and display the real-time data collected from the incubator's sensors. This application provides a user-friendly

interface for healthcare professionals to monitor the infant's vital signs and environmental conditions remotely.

Real-time Data Visualization:

The collected data is visualized in the application's user interface, typically in the form of charts, graphs, or numeric values. These visual representations provide a clear and concise overview of the infant's current condition.

Immediate Alerts and Notifications:

The real-time monitoring system is capable of detecting abnormal or critical changes in the infant's vital signs or environmental conditions. If any parameter exceeds predetermined thresholds or falls outside the normal range, the system generates immediate alerts or notifications. These alerts can be sent to healthcare professionals' devices, such as smartphones or computers, ensuring prompt attention and intervention.

Remote Access and Monitoring:

Healthcare professionals can access the real-time data and monitor the infant's condition remotely, regardless of their location. This feature enables healthcare providers to
provide timely care and make informed decisions promptly, even if they are not physically present in the neonatal care unit.

Historical Data Comparison:

In addition to real-time monitoring, the system may also store historical data for reference and analysis. This allows healthcare professionals to compare current data with previous readings, identifying trends or patterns that may be relevant for diagnosis, treatment, or evaluation of the infant's progress.

4.4 Automated alerts and notifications

Abnormal Condition Detection:

The smart baby incubator continuously monitors the infant's vital signs and environmental conditions through sensors. The collected data is analyzed in real-time to detect any abnormal conditions or deviations from the normal range. For example, if the temperature or oxygen level exceeds a predefined threshold or falls below a critical level, it indicates a potential risk to the infant's health.

Rule-Based Alert Generation:

Based on predefined rules and thresholds, the smart baby incubator system generates automated alerts when an abnormal condition is detected. These rules can be set according to medical guidelines or individual patient requirements. For instance, if the temperature rises above a specific threshold for a certain duration, an alert is triggered.

Alert Transmission:

Once an alert is generated, the system uses various communication channels to transmit the alert to the appropriate recipients. This can include sending notifications to healthcare professionals' mobile devices, email, or pager systems. The chosen communication
method should ensure that the alerts are promptly delivered and received.

Clear and Actionable Information:

The automated alerts should provide clear and concise information about the detected abnormality or critical condition. The alert should include relevant details such as the type of abnormality, its severity, and any recommended actions or interventions. This helps healthcare professionals or caregivers to understand the situation quickly and take appropriate actions.

Timely Delivery:

Timeliness is crucial when it comes to alerting healthcare professionals about critical conditions in a smart baby incubator. The system should transmit the alerts in real-time or near real-time to ensure that healthcare providers receive the information promptly. Delays in alert delivery could have adverse consequences for the infant's well-being.

Acknowledgment and Response:

The smart baby incubator system may include a mechanism for healthcare professionals to acknowledge or respond to the alerts. This acknowledgment can serve as confirmation that the alert has been received and that appropriate actions are being taken.
Additionally, the system may provide a log or record of the alerts and responses for documentation and analysis.

Customization and Priority Levels:

The system should allow customization of alert settings based on individual patient needs or medical protocols. Different alert priority levels can be assigned to ensure that

critical conditions receive immediate attention, while non-urgent alerts are appropriately managed.

4.5 Customized Incubator Environment

Remote Control:

The IoT-enabled smart baby incubator allows healthcare professionals to remotely control and adjust the incubator's environmental factors. This control can be accessed through a web or mobile application.

Temperature Control:

Temperature is a critical factor in maintaining the well-being of the infant. The smart baby incubator allows for remote adjustment of the incubator's temperature based on the specific needs of the infant. For example, if the infant requires a higher temperature to promote growth or comfort, healthcare professionals can increase the temperature remotely.

Humidity Control:

Humidity plays a vital role in maintaining the moisture levels necessary for the infant's skin and respiratory health. The smart baby incubator enables healthcare professionals to remotely control the humidity levels within the incubator. This feature ensures that the infant is exposed to an appropriate level of humidity to prevent dryness or excessive moisture.

Oxygen Control:

The smart baby incubator system also allows for remote control of the oxygen levels within the incubator. Healthcare professionals can adjust the oxygen concentration based on the infant's specific needs. This feature ensures that the infant receives the optimal amount of oxygen required for their respiratory health.

Adaptive Control:

In addition to manual adjustments, the smart baby incubator using IoT can employ adaptive control mechanisms. These mechanisms utilize algorithms and machine learning to continuously monitor the infant's vital signs and adjust the incubator's environmental factors automatically. This adaptive control ensures that the incubator environment is dynamically optimized to suit the changing needs of the infant.

Personalized Settings:

Each infant may have specific requirements or conditions that need to be considered. The smart baby incubator system allows healthcare professionals to create personalized settings for each individual infant. These settings can be saved and easily applied when needed, ensuring that the incubator environment is tailored to meet the specific needs of each baby.

4.6 Enhanced safety and security

Data Encryption:

The smart baby incubator system incorporates data encryption techniques to protect the confidentiality and integrity of the collected data. Encryption algorithms are used to encode the data during transmission and storage, making it unreadable to unauthorized individuals or malicious attackers. This ensures that sensitive information about the infant's vital signs and health remains secure.

Access Control:

Access control mechanisms are employed to restrict unauthorized access to the smart baby incubator system. Authentication methods such as username/password combinations, biometric authentication, or two-factor authentication can be implemented to ensure that only authorized healthcare professionals can access and control the incubator system. This prevents unauthorized individuals from tampering with the incubator's settings or accessing sensitive data.

Secure Communication:

The smart baby incubator system employs secure communication protocols to safeguard the transmission of data between the incubator and the cloud-based platform or local server. Secure socket layers (SSL) or transport layer security (TLS) protocols can be implemented to establish encrypted connections, preventing unauthorized interception or tampering of data during transmission.

Physical Security:

Physical security measures are implemented to ensure the physical safety of the smart baby incubator. These measures may include restricted access to the incubator unit, surveillance systems, and monitoring to prevent unauthorized physical tampering or

access.

Threat Monitoring and Detection:

The smart baby incubator system can incorporate threat monitoring and detection mechanisms to identify potential security breaches or anomalies. Intrusion detection systems (IDS) or anomaly detection algorithms can be used to continuously monitor the system for any suspicious activities or unauthorized access attempts. In case of any detected threats, appropriate alerts or notifications can be generated for immediate action.

Regular Software Updates and Patching:

To address any potential security vulnerabilities, regular software updates and patching are crucial. The smart baby incubator system should be designed to receive and install necessary updates to the software and firmware components. This ensures that the system remains up-to-date with the latest security patches, minimizing the risk of exploitation by malicious actors.

Compliance with Privacy Regulations:

The smart baby incubator system must comply with applicable privacy regulations, such as General Data Protection Regulation (GDPR) or Health Insurance Portability and Accountability Act (HIPAA). Compliance ensures that the collection, storage, and processing of the infant's data adhere to the necessary privacy and security standard.

CHAPTER-5

HARDWARE AND SOFTWARE IMPLEMENTATION

Hardware and software implementation for a smart baby incubator using IoT, where the data from sensors is sent to a website made with HTML, CSS, JS, and Arduino IoT Cloud for the dashboard, involves both the physical components and the software systems that enable the functionality. Here is a detailed explanation of hardware and software implementation in this context:

Hardware Implementation:

1. Sensors: The smart baby incubator incorporates various sensors such as temperature, humidity, oxygen level sensors, etc. These sensors are connected to the Arduino microcontroller board to collect data on the infant's vital signs and environmental conditions.

2. Arduino Microcontroller Board: An Arduino board acts as the brain of the smart baby incubator system. It interfaces with the sensors to collect the sensor data and communicates with the ESP01 module for data transmission.

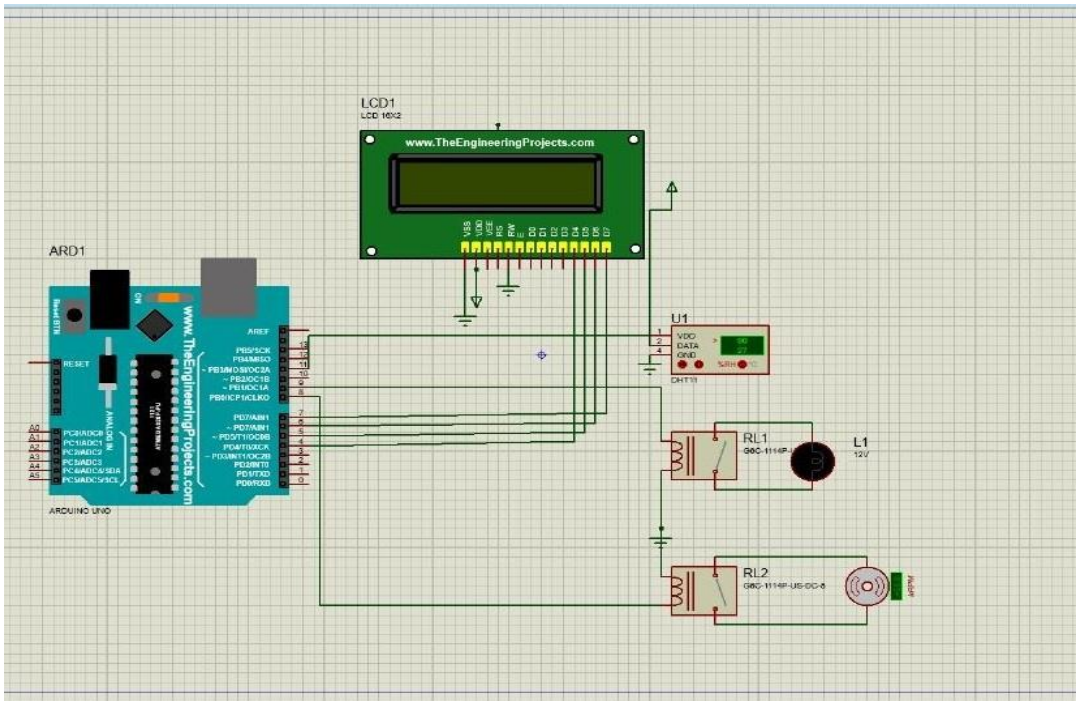


Fig 2. Circuit Diagram

3. ESP01 Module: The ESP01 module is a Wi-Fi-enabled microcontroller module that enables the smart baby incubator to establish a wireless internet connection. It is responsible for transmitting the collected sensor data to the Arduino IoT Cloud.

4. LCD Module: LCD 16*2 refers to a type of liquid crystal display (LCD) that has 16 characters per line and 2 lines. They provide a visual interface for displaying information in a clear and readable format. To control an LCD 16*2, you would typically use a microcontroller or other digital device that sends commands and data to the display through a communication protocol such as I2C or SPI. The microcontroller can then instruct the LCD to display specific characters or strings of text, change the cursor position, or control other features like scrolling or blinking.

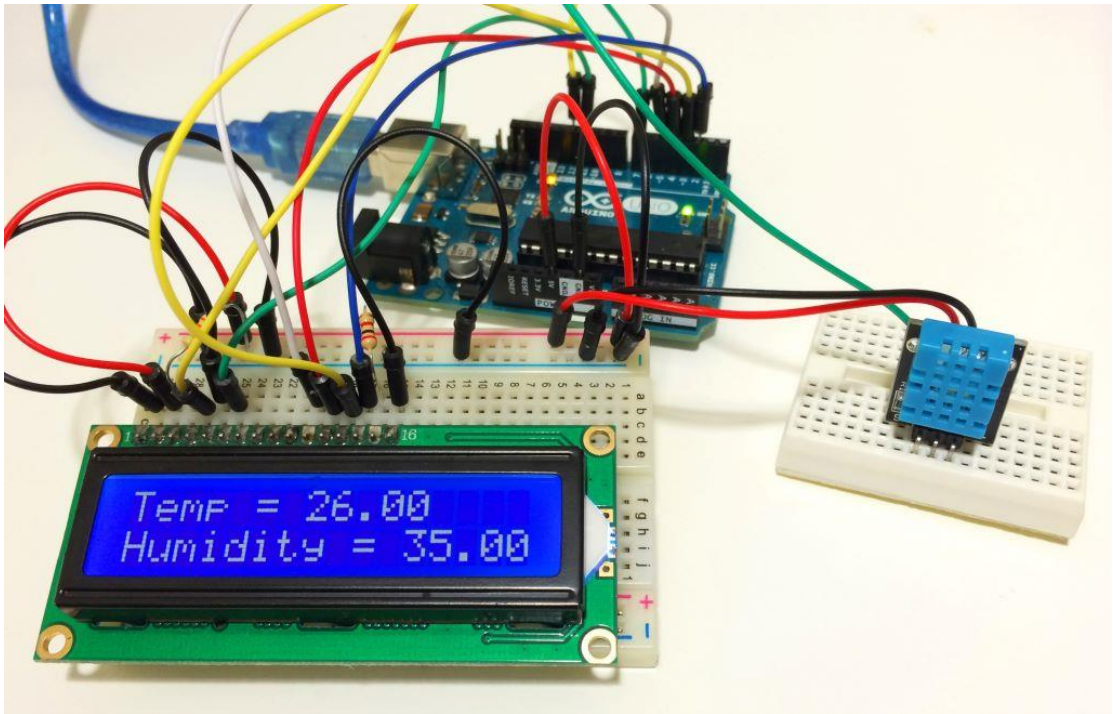


Fig 3. Arduino connection with LCD and DHT11 sensor

5. Power Supply: The hardware components of the smart baby incubator, including sensors, Arduino board, and ESP01 module, require a stable power supply. This can be provided by batteries or an external power source.

Software Implementation:

1. Arduino Firmware Development: The firmware is the software that runs on the Arduino microcontroller board. It is responsible for reading data from the sensors, processing it, and sending it to the Arduino IoT Cloud. The firmware needs to be programmed to handle the sensor data and prepare it for transmission.

2. Wi-Fi Configuration: The ESP01 module needs to be configured to connect to the Wi-Fi network, including providing the necessary network credentials (such as SSID and password) in the firmware code. This ensures the ESP01 can establish a connection to the internet and communicate with the Arduino IoT Cloud.

3. Data Transmission to Arduino IoT Cloud: The firmware is programmed to send the collected sensor data to the Arduino IoT Cloud. The Arduino IoT Cloud provides an online platform that allows you to create a dashboard for monitoring and visualizing the real-
time data. The firmware needs to establish a secure connection with the Arduino IoT Cloud and transmit the data using the provided APIs or protocols.

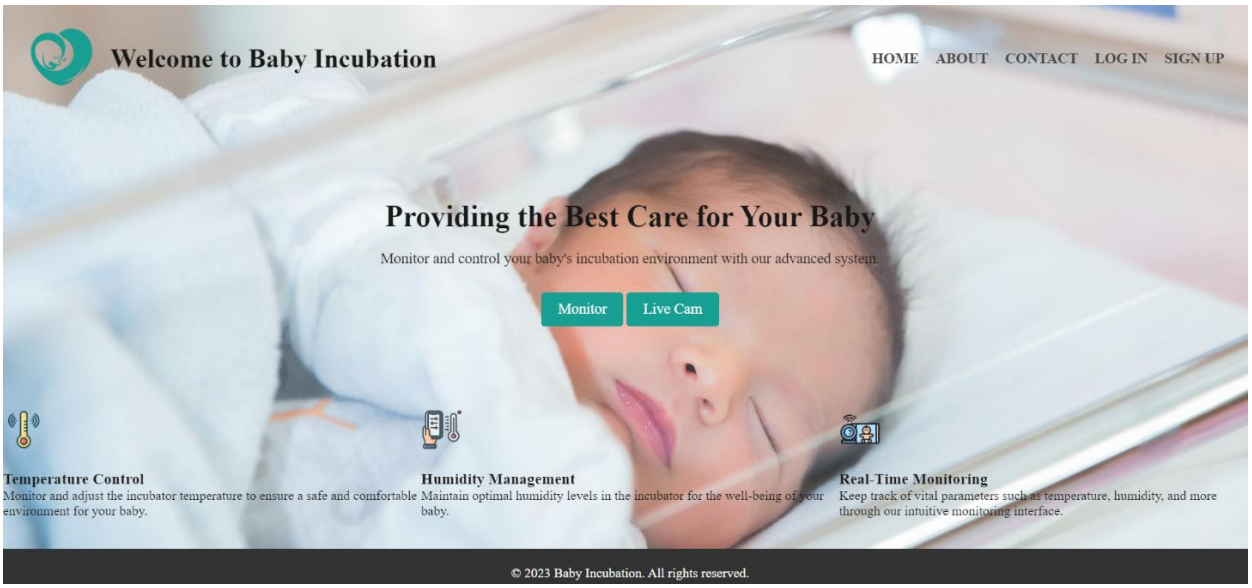


Fig 4. Website Homepage

4. Web Development: On the website side, you will use HTML, CSS, and JavaScript (JS) to create the web pages and functionality. HTML provides the structure of the web page, CSS handles the styling and layout, and JS enables interactivity and dynamic content. The website can be used to display the real-time data from the Arduino IoT Cloud dashboard.

5. Real-time Data Visualization: The Arduino IoT Cloud provides tools and APIs to create a dashboard for real-time data visualization. Using HTML, CSS, and JS, you can embed or integrate the Arduino IoT Cloud dashboard into the website. This allows for dynamic updates and real-time visualization of the sensor data on the website.

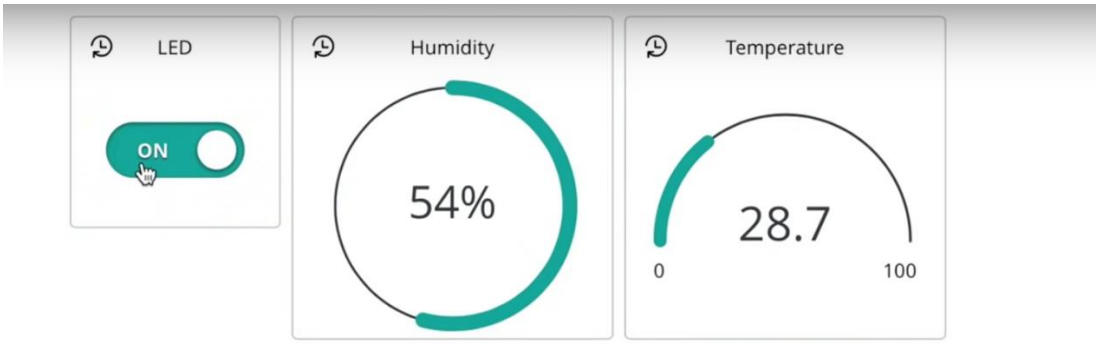


Fig 5. Dashboard

6. Security Measures: Implementing security measures is crucial to protect the privacy and integrity of the data. This can include encryption for secure data transmission, user authentication to access the website, and input validation on the server-side to prevent malicious actions.

   By implementing the hardware and software components mentioned above, the smart baby incubator system can collect sensor data, transmit it to the Arduino IoT Cloud in real-time using ESP01, and display it on a website made with HTML, CSS, and JS. The hardware components gather the sensor data, while the firmware and web application handle the data processing, transmission, and visualization. The Arduino IoT Cloud provides the platform for monitoring and visualizing the real-time data through its dashboard functionality.

CHAPTER -6

RESULTS AND DISCUSSION

A. Temperature and Humidity monitoring

Temperature: 30.00 degrees Celsius, Humidity: 38.00%.	Temperature: 37.00 degrees Celsius, Humidity: 34.00%.
Temperature: 30.00 degrees Celsius, Humidity: 31.00%.	Temperature: 37.00 degrees Celsius, Humidity: 36.00%.
Temperature: 30.00 degrees Celsius, Humidity: 31.00%.	Temperature: 37.00 degrees Celsius, Humidity: 34.00%.
Temperature: 30.00 degrees Celsius, Humidity: 39.00%.	Temperature: 37.00 degrees Celsius, Humidity: 33.00%.
Temperature: 30.00 degrees Celsius, Humidity: 38.00%.	Temperature: 37.00 degrees Celsius, Humidity: 33.00%.
Temperature: 30.00 degrees Celsius, Humidity: 37.00%.	Temperature: 37.00 degrees Celsius, Humidity: 32.00%.
Temperature: 30.00 degrees Celsius, Humidity: 32.00%.	Temperature: 37.00 degrees Celsius, Humidity: 31.00%.
Temperature: 30.00 degrees Celsius, Humidity: 33.00%.	Temperature: 35.00 degrees Celsius, Humidity: 30.00%.
Temperature: 31.00 degrees Celsius, Humidity: 33.00%.	Temperature: 34.00 degrees Celsius, Humidity: 35.00%.
Temperature: 31.00 degrees Celsius, Humidity: 37.00%.	Temperature: 34.00 degrees Celsius, Humidity: 30.00%.
Temperature: 33.00 degrees Celsius, Humidity: 35.00%.	Temperature: 34.00 degrees Celsius, Humidity: 29.00%.
Temperature: 34.00 degrees Celsius, Humidity: 36.00%.	Temperature: 34.00 degrees Celsius, Humidity: 29.00%.
Temperature: 36.00 degrees Celsius, Humidity: 37.00%.	Temperature: 35.00 degrees Celsius, Humidity: 29.00%.

B. Serial Monitor

```
D!Dypp`ε`!   ****** Arduino IoT Cloud - configuration info *****
Device ID: 163a557c-237d-44fd-ad3a-30010a46a25c
Thing ID: 62087188-8f85-40fa-8f7a-89c4d341d379
MQTT Broker: mqtts-up.iot.arduino.cc:8884
WiFi status ESP: 255
Temperature - 28.70
Humidity - 56.00
Connected to "SmS_jiofi"
Temperature - 28.70
Humidity - 55.00
Temperature - 28.70
Humidity - 55.00
Temperature - 28.70
Humidity - 56.00
Connected to Arduino IoT Cloud
Temperature - 28.70
Humidity - 55.00
Temperature - 28.70
Humidity - 55.00
```

Fig 6. Serial Monitor

CHAPTER -7

CONCLUSION

The proposed system monitors Oxygen levels of the infant and temperature and humidity of the surrounding. Temperature monitoring is done in order to keep the environment suitable for the neonate. Temperature monitoring of the infant's body will help to detect many other internal diseases like infections, common cold, and pneumonia, have a common symptom of fever as the body temperature goes high. Humidity measure values also help in the detection of having internal problems like cold, dehydration. But the temperature inside the incubator loose due to atmosphere or any other problems, the heating pad will automatically on. And also, for continuous monitoring there should always have the facilities of power supply. But in many places, there is still have the problem of power cut about several hours.

7.1 Future scope

The future work is focused on the implementation of the monitoring system to monitor all the pathological parameters of infants in all the hospitals with common physician/consultant. This paper aim is to monitoring and controlling temperature within the incubator and checking the infant's heartbeat continuously. Oral temperature will
show a time dependent increase in value after capture; heart rate and respiration rate will stabilize over time. A positive correlation exists between heart rate, potassium, and lactate concentration amongst healthy captured manatees. In future, many implementations can take place, like ECG monitoring, proper oxygen supply controlling and above all the controlling system will automatically on or off. As there will be given a particular set point for different parameters, the controller will automatically on if the measuring value goes below the set value and the controller will automatically off if the measuring value goes high compare to the set value. There is also a possibility of replace the power source by solar cell. In rural area there is a big problem in supplying of electricity. So, this problem
will also overcome in future implementations.

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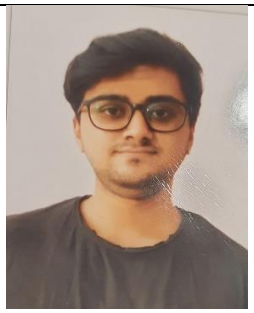
1	Mohammed Saqib	1BY20EC101		1by20ec101@bmsit.in	8217362970
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